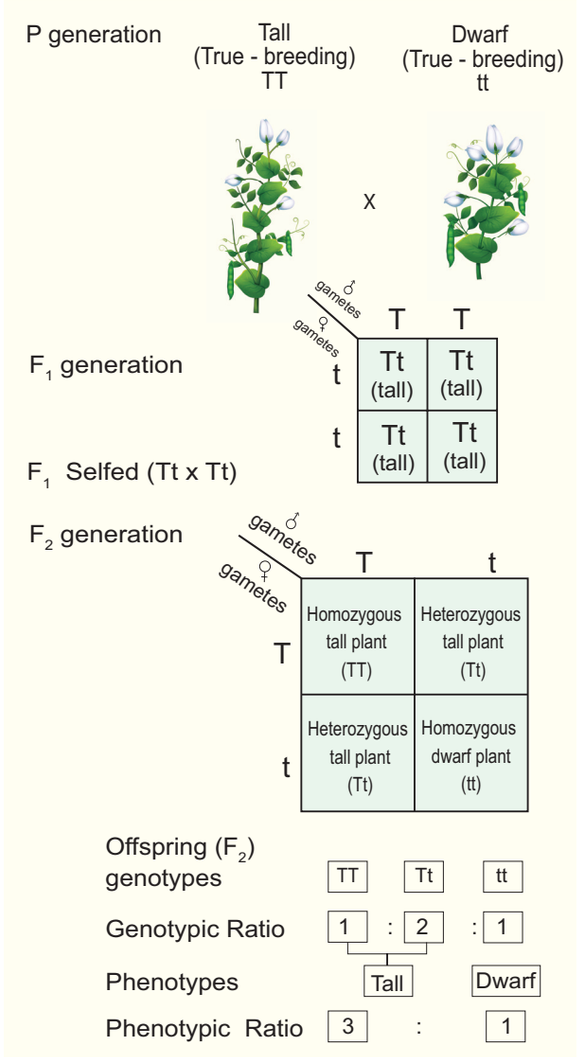


Figure 2.6: Monohybrid Cross

2.3.1 Mendel's analytical and empirical approach

Mendel chose two contrasting traits for each character. So it seemed logical that two distinct factors exist. In F_1 the recessive trait and its factors do not disappear and they are hidden or masked only to reappear in ¼ of the F_2 generation. He concluded that tall and dwarf alleles of F_1 heterozygote segregate randomly into gametes. Mendel got 3:1 ratio in F_2 between the dominant and recessive trait. He was the first scientist to use this type of quantitative analysis in a biological experiment. Mendel's data is concerned with the proportions of offspring.

Mendel's analytical approach is truly an outstanding scientific achievement. His meticulous work and precisely executed breeding experiments proposed that discrete particulate units of heredity are present and they are transmitted from one generation to the other. Now they are called as genes. Mendel's experiments were well planned to determine the relationships which govern hereditary traits. This rationale is called an empirical approach. Laws that were arrived from an empirical approach is known as empirical laws.

2.3.2 Test cross

Test cross is crossing an individual of dominant phenotype but unknown genotype with a homozygous recessive.

In Mendel's monohybrid cross all the plants are tall in F_1 generation. In F_2 tall and dwarf plants were in the ratio of 3:1. Mendel self pollinated dwarf F_2 plants and got dwarf plants in F_3 and F_4 generations. So he concluded that the genotype of dwarf was homozygous (tt). The genotypes of tall plants TT or Tt from $F_1$and F_2 cannot be predicted. But how we can tell if a tall plant is homozygous or heterozygous? To determine the genotype of a tall plant Mendel crossed the plants from F_2 with the homozygous recessive dwarf plant. This he called a test cross. The progenies of the test cross can be easily analysed to predict the genotype of the plant or the test organism. Thus in a typical test cross an organism (pea plants) showing dominant phenotype (whose genotype is to be determined) is crossed with the recessive parent instead of self crossing. Test cross is used to identify whether an individual is homozygous or heterozygous for dominant character.

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